

Preprocessing Steps and Business Questions with Visualization

Date	02 July 2026
Team ID	SWTID-2026-5525
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	

1. Project Overview

The Electric Cars Analytics project focuses on analyzing electric vehicle datasets using MySQL, SQL, Tableau and Flask integration. The system processes EV-related information such as vehicle specifications, charging stations, pricing, efficiency, and brand performance to generate interactive analytical dashboards.

2. Dataset Collection

Datasets Used:

1. ElectricCarData_Clean.csv

Description:

- EV brand information
- Vehicle models
- Battery range
- Top speed
- Efficiency
- Pricing details

2. EVIndia.csv

Description:

- Indian electric vehicle data
- EV models
- Vehicle categories

3. Electric Vehicle Charging Station List.csv

Description:

- Charging station locations
- State/Region details
- Latitude and Longitude information

4. Cheapest Electric Cars Database.csv

Description:

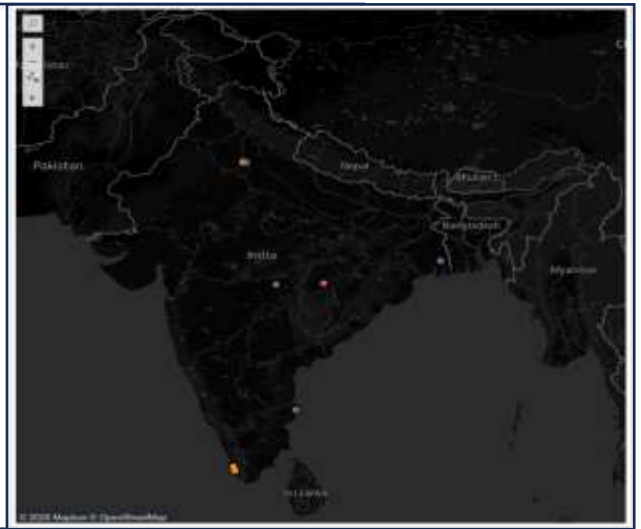
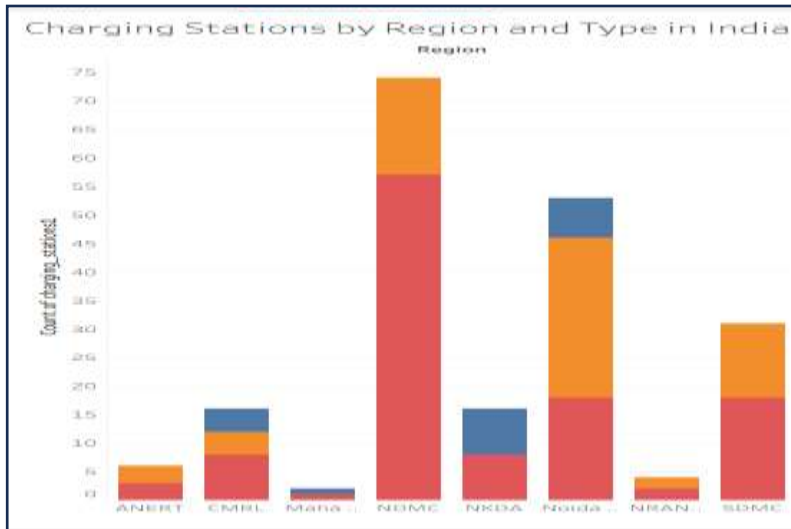
- Global EV pricing
- Vehicle performance details

3. Business Questions with Visualization

Business Question 1: How is the EV charging infrastructure distributed across different regions?

Visualization:

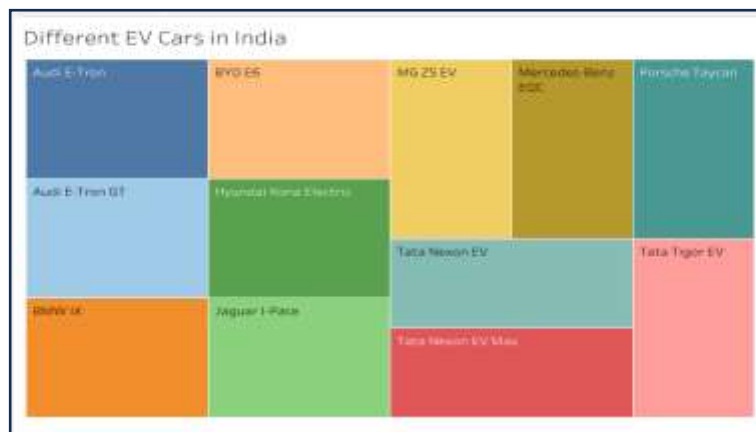
- Charging Stations by Region
- EV Charging Map



Business Question 2: Which electric vehicle brands and models dominate the EV market?

Visualization Used:

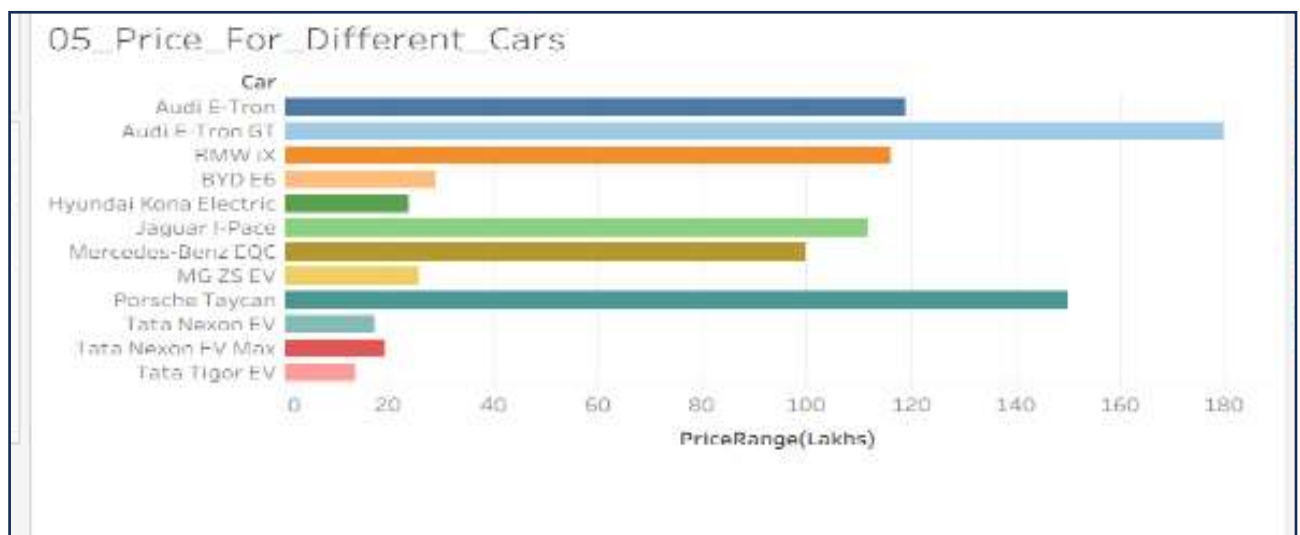
- Different EV Cars (Treemap)



Business Question 3: How does the price range vary among different electric vehicle brands?

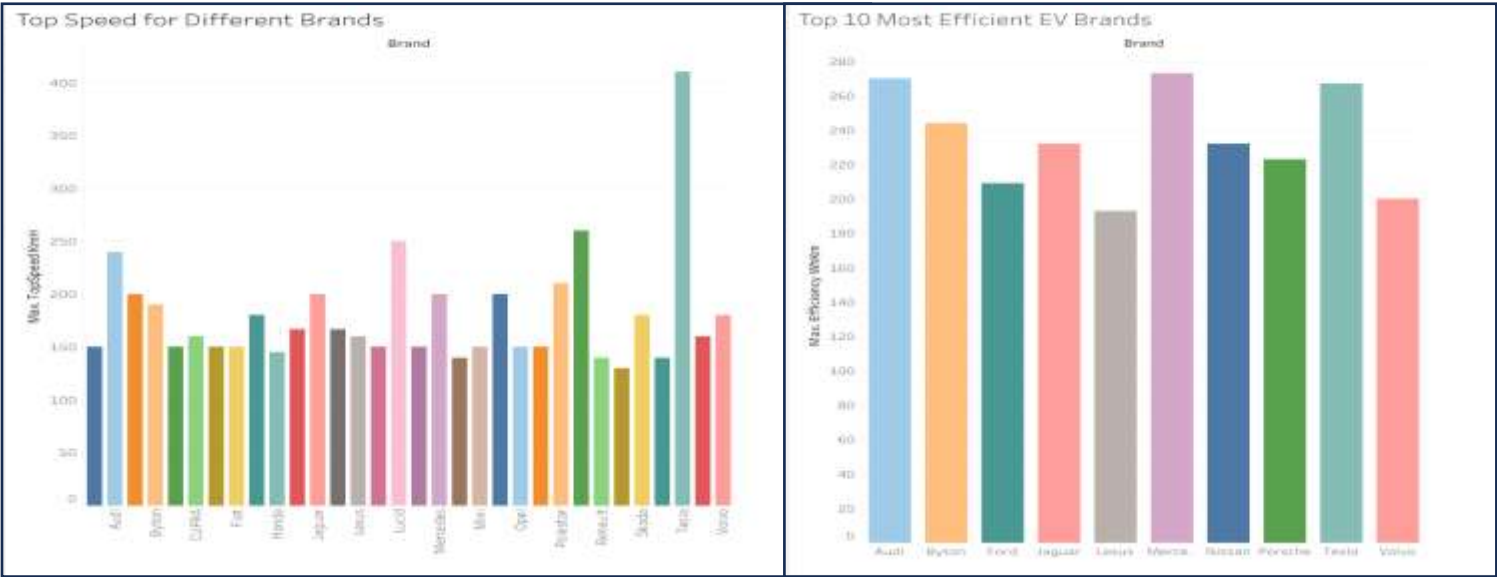
Visualization Used:

- Price for Different Cars



Business Question 4: Which EV brands provide better performance based on speed and efficiency?
Visualization Used:

- Top Speed
- Top 10 Efficient Brands



Business Question 5: How are EVs classified based on design and powertrain technology?
Visualization Used:

- Brands by BodyStyle
- Brand Filtered by PowerTrain

